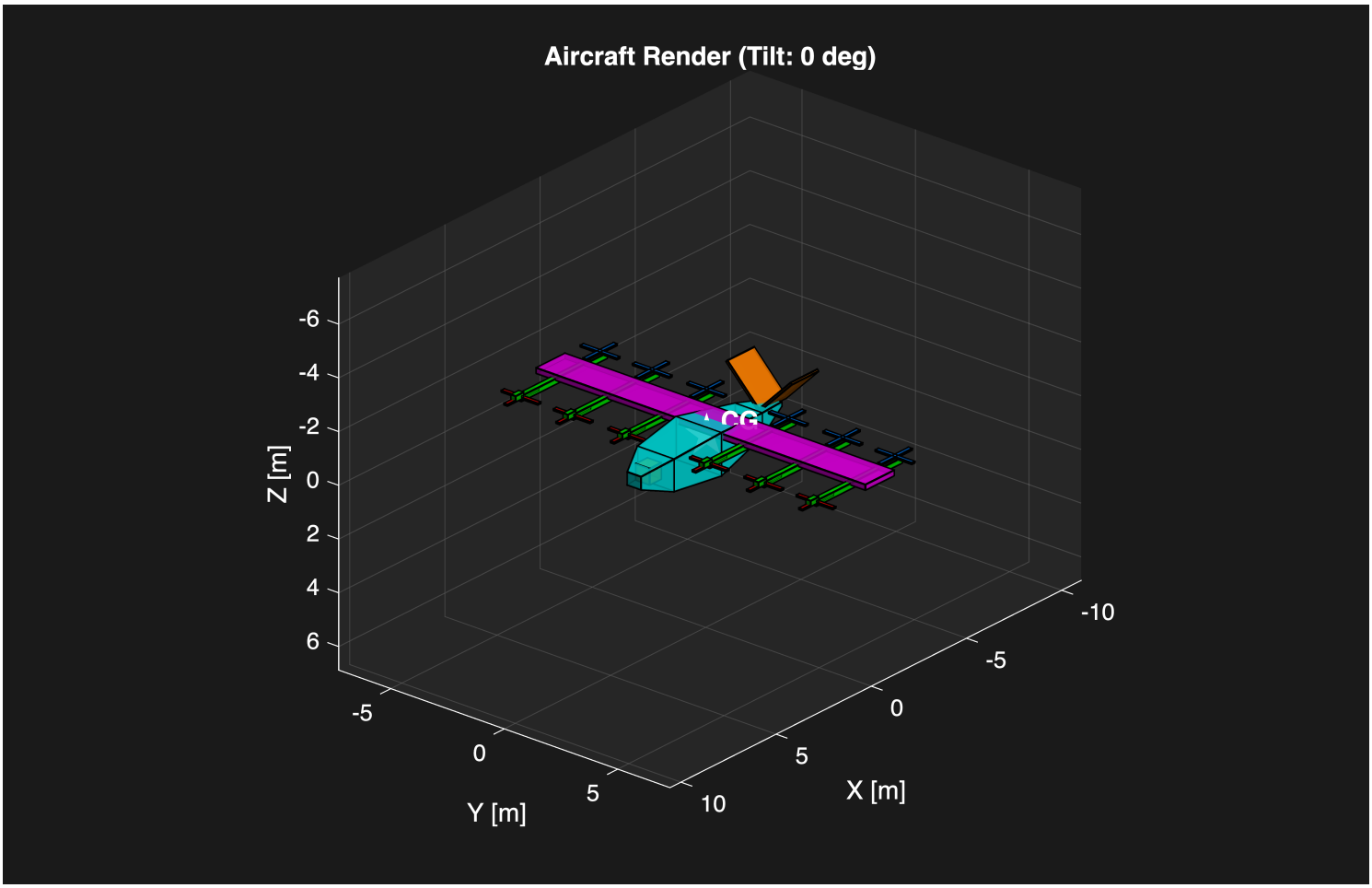
```
clear; close all; clc;

tester_model = 'TESTER_Brown_Full_Sim';
full_model = 'Brown_Full_Sim';
disp('Model is significantly differnt from given model (see above)/n many
test case parameters had to be ajusted')
```

```
Model is significantly differnt from given model (see above)/n many test case parameters had to be ajusted
```

```
[Mass, CG, J, wing, tailL, tailR, prop, compData, tilt_angle] =
init_from_full_sim();
render_aircraft(compData, CG, tilt_angle);
```



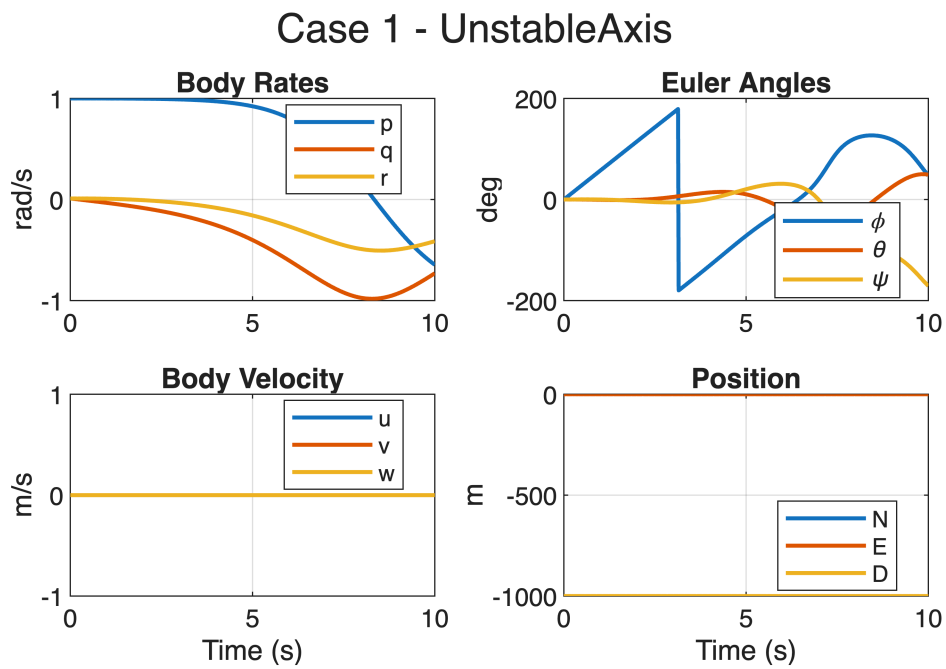

```
Motor_RPMs = zeros(12,1);
Tilt_angles = zeros(6,1);

%% Test cases


TC1.name = 'Case 1 - UnstableAxis';
TC1.model = tester_model;
```

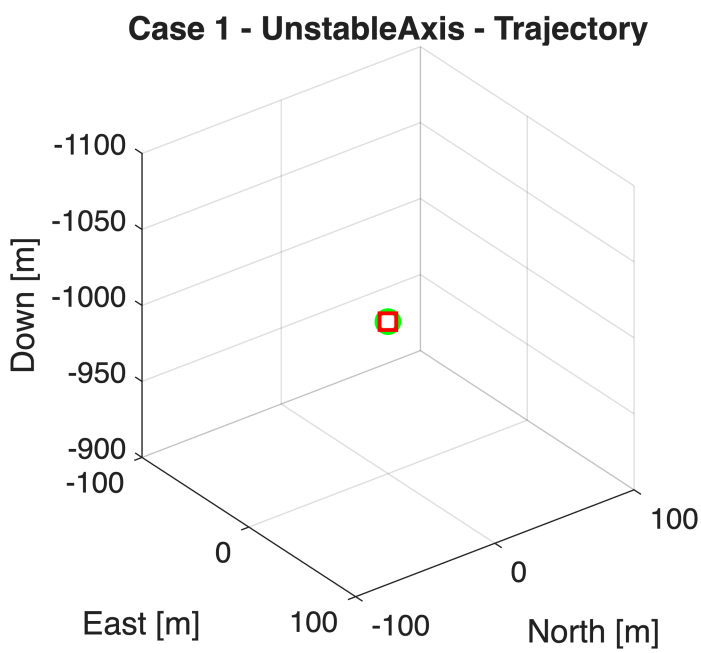
```
TC1.t_stop = 10;
TC1.pos_init = [0;0;-1000];
TC1.V_init = [0;0;0];
TC1.eul_init_deg = [0;0;0];
TC1.omega_init = [1;1e-2;1e-2];
TC1.g = 0;
TC1.Fext_B = [0;0;0];
TC1.Mext_B = [0;0;0];
TC1.Motor_RPMs = Motor_RPMs;
TC1.Tilt_angles = Tilt_angles;
TC1.note = 'Spins about axis and couples into motion in other axes as
expected';

TC2.name = 'Case 2 - OutsideLoop';
TC2.model = tester_model;
TC2.t_stop = 10;
TC2.pos_init = [0;0;-1000];
TC2.V_init = [13;0;0];
TC2.eul_init_deg = [0;0;0];
TC2.omega_init = [0;1;0];
TC2.g = 0;
TC2.Fext_B = [0;0;-70000];
TC2.Mext_B = [0;0;0];
TC2.Motor_RPMs = Motor_RPMs;
TC2.Tilt_angles = Tilt_angles;
TC2.note = 'DLoops as expected, not a perfect circle/nbecause I didnt
calculate the right force';

TC3.name = 'Case 3 - ForceField';
TC3.model = tester_model;
TC3.t_stop = 10;
TC3.pos_init = [0;0;-1000];
TC3.V_init = [13;0;0.25];
TC3.eul_init_deg = [0;0;0];
TC3.omega_init = [0;0;0];
TC3.g = 9.81;    %gravity changed to positive to make vector pointing down
TC3.Fext_B = [0;0;-4.7088];
TC3.Mext_B = [0;0;0];
TC3.Motor_RPMs = Motor_RPMs;
TC3.Tilt_angles = Tilt_angles;
TC3.note = 'Drops due to gravity being stronger than the external force';

TC4.name = 'Case 4 - Roll Damping';
TC4.model = full_model;
TC4.t_stop = 10;
TC4.pos_init = [0;0;-1000];
TC4.V_init = [250;0;0];
TC4.eul_init_deg = [0;0;0];
TC4.omega_init = [0.3;0;0];
TC4.g = [0;0;0];
```

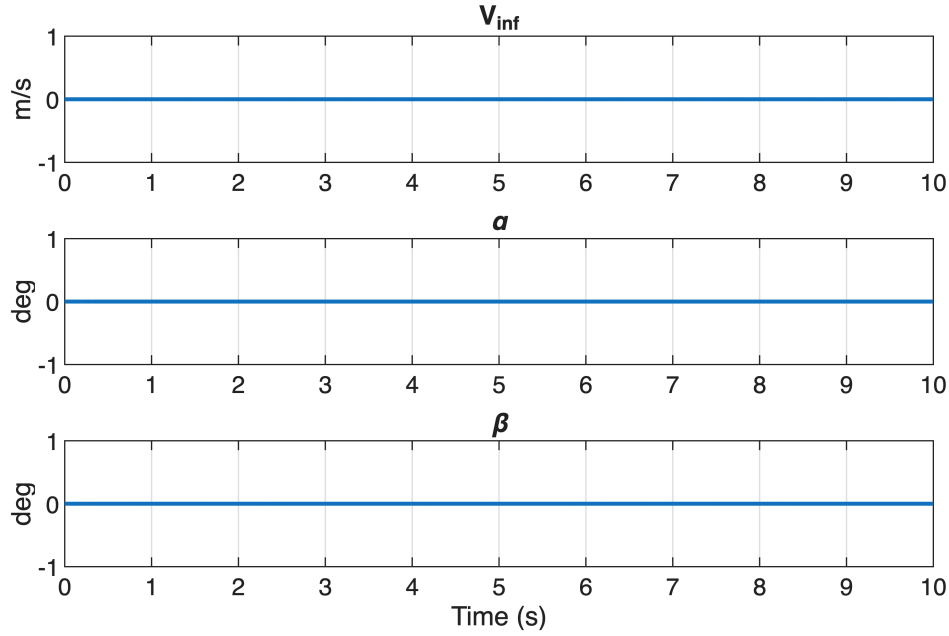
```
TC4.Fext_B = [0;0;0];
TC4.Mext_B = [0;0;0];
TC4.Motor_RPMs = Motor_RPMs;
TC4.Tilt_angles = Tilt_angles;
TC4.note = 'Stability seems to be different from prop plane likely due to
V tail /nSeems like ti has a stability simmilar to a paper airplane that/n
speeds while dropping then pitches and slows, before dropping again';

TC5.name = 'Case 5 - Beta to Yaw';
TC5.model = full_model;
TC5.t_stop = 10;
TC5.pos_init = [0;0;-1000];
TC5.V_init = [250;25;0];
TC5.eul_init_deg = [0;0;0];
TC5.omega_init = [0;0;0];
TC5.g = [0;0;0];
TC5.Fext_B = [0;0;0];
TC5.Mext_B = [0;0;0];
TC5.Motor_RPMs = Motor_RPMs;
TC5.Tilt_angles = Tilt_angles;
TC5.note = 'again lots of coupling in the axes that cant be simply
explained due to V tial/n seeing the same paper airplane synamics here';

TC6.name = 'Final Validation - Stable 6DOF';
TC6.model = full_model;
TC6.t_stop = 10;
TC6.pos_init = [0;0;-1000];
TC6.V_init = [200;0.25;0.25];
TC6.eul_init_deg = [10;10;0];
TC6.omega_init = [1e-2;1;1e-2];
TC6.g = [0;0;-9.81];
TC6.Fext_B = [0;0;0];
TC6.Mext_B = [0;0;0];
TC6.Motor_RPMs = 1500 * ones(12,1);
TC6.Tilt_angles = 90 * ones(6,1);
TC6.note = 'Motors seem to bring plane upwards - Aircraft is much more
complex than simple prop plane/n Will take a long time to develop realistic
test cases';

run_flight_test(TC1, Mass, CG, J, wing, tailL, tailR, prop);
```

```
--------------------------------------------------
Case 1 - UnstableAxis
--------------------------------------------------
  Model:            TESTER_Brown_Full_Sim
  Simulation Time:  10 s
  Initial Position: [0  0  -1000] m
  Initial Velocity: [0  0  0] m/s
  Initial Euler:    [0  0  0] deg
  Initial Omega:    [1  0.01  0.01] rad/s
  Gravity:          [0  0  0] m/s^2
  Fext_B:           [0  0  0] N
```

```
Mext_B:            [0  0  0] N-m
```


Case 1 - UnstableAxis
Body Rates
rad/s
p
q
r
Euler Angles
deg
ϕ
θ
ψ
Body Velocity
m/s
u
v
w
Time (s)
Position
m
N
E
D
Time (s)

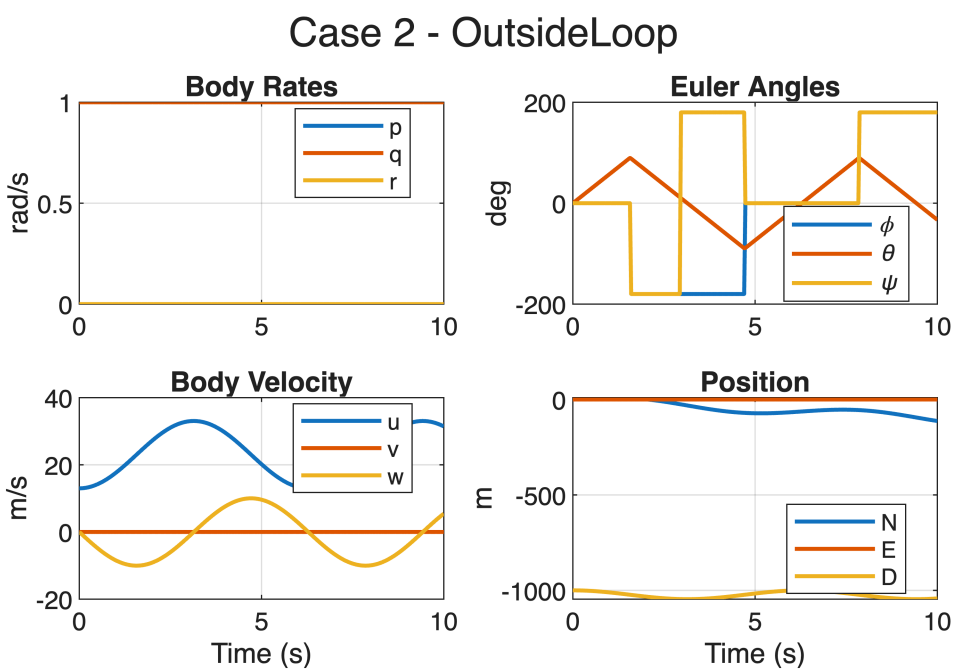

Case 1 - UnstableAxis - Trajectory
Down [m]
East [m]
North [m]

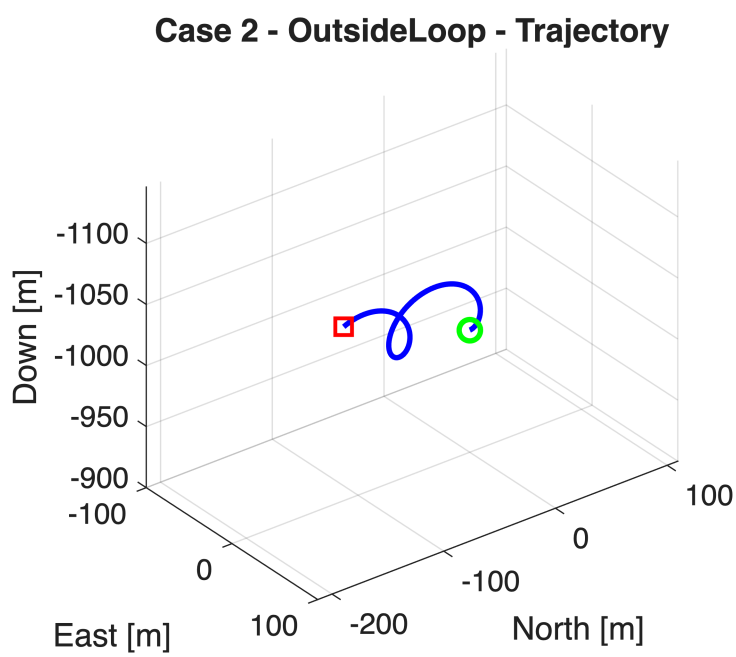


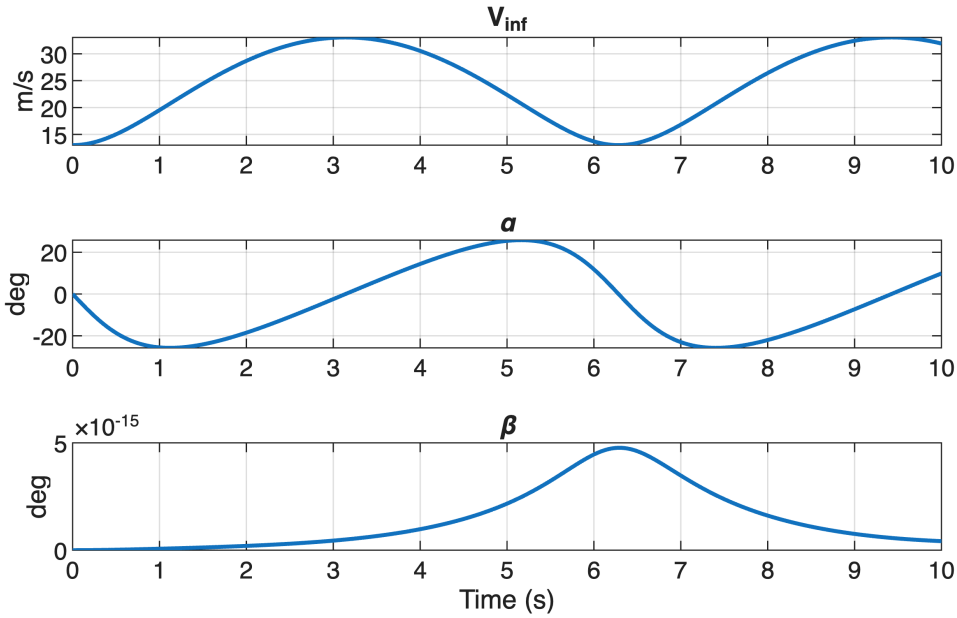
Spins about axis and couples into motion in other axes as expected

```
out = run_flight_test(TC2, Mass, CG, J, wing, tailL, tailR, prop);
```

```
--------------------------------------------------
Case 2 - OutsideLoop
--------------------------------------------------
  Model:             TESTER_Brown_Full_Sim
  Simulation Time:  10 s
  Initial Position: [0  0  -1000] m
  Initial Velocity: [13  0  0] m/s
  Initial Euler:    [0  0  0] deg
  Initial Omega:    [0  1  0] rad/s
  Gravity:          [0  0  0] m/s^2
  Fext_B:           [0  0  -70000] N
  Mext_B:           [0  0  0] N-m
```


Case 2 - OutsideLoop
Body Rates
rad/s
p
q
r
Euler Angles
deg
ϕ
θ
ψ
Body Velocity
m/s
u
v
w
Time (s)
Position
m
N
E
D
Time (s)

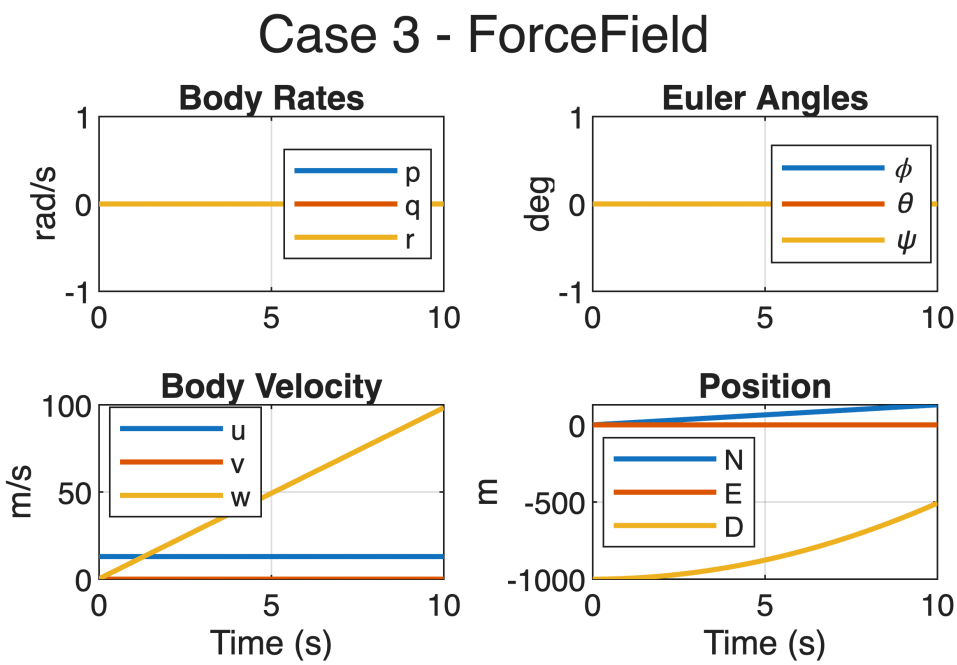

Case 2 - OutsideLoop - Trajectory
Down [m]
East [m]
North [m]

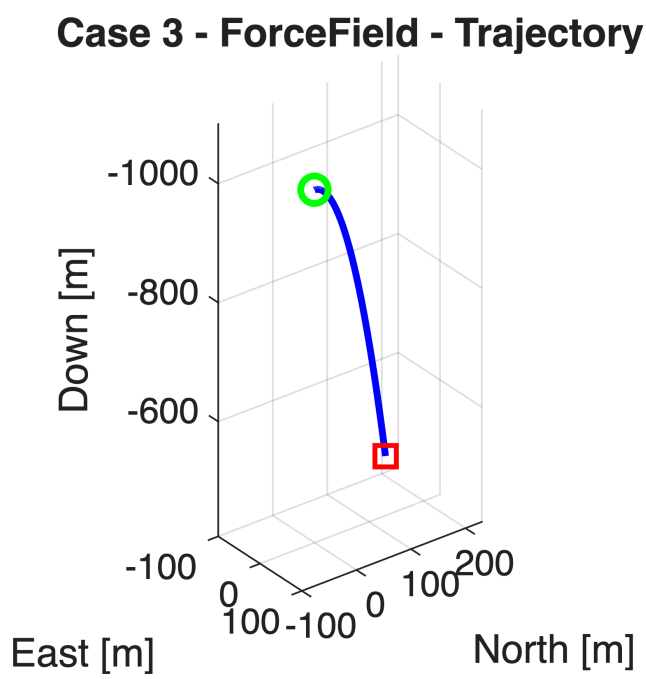


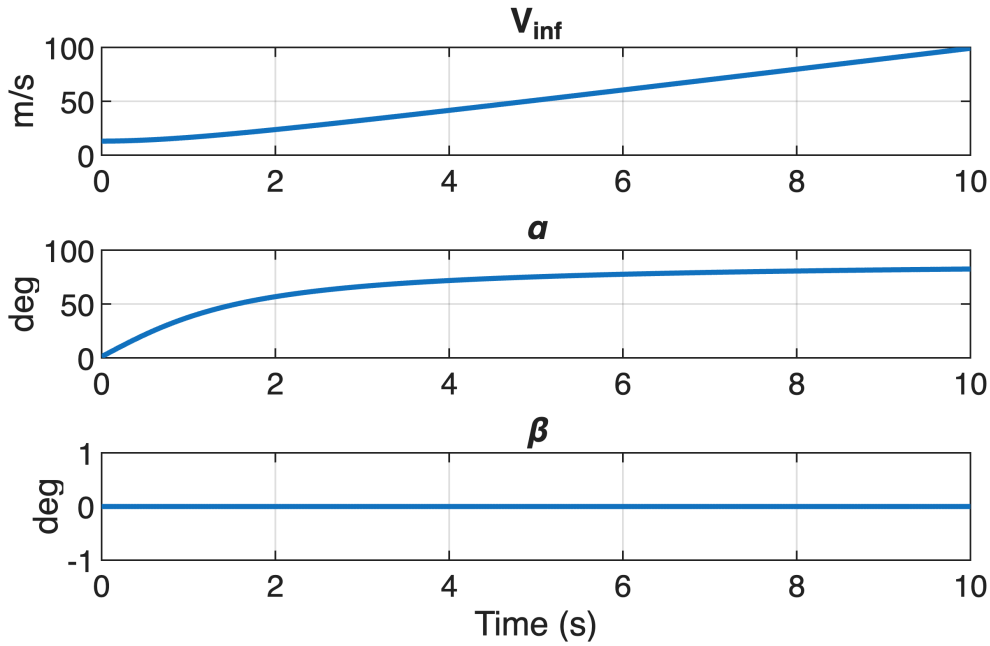
DLoops as expected, not a perfect circle/nbecause I didnt calculate the right force

```
run_flight_test(TC3, Mass, CG, J, wing, tailL, tailR, prop);
```

```
----------------------------------------------------
Case 3 - ForceField
----------------------------------------------------
  Model:             TESTER_Brown_Full_Sim
  Simulation Time:   10 s
  Initial Position:  [0  0  -1000] m
  Initial Velocity:  [13  0  0.25] m/s
  Initial Euler:     [0  0  0] deg
  Initial Omega:     [0  0  0] rad/s
  Gravity:           [0  0  9.81] m/s^2
  Fext_B:            [0  0  -4.7088] N
  Mext_B:            [0  0  0] N-m
```


Case 3 - ForceField
Body Rates
rad/s
p
q
r
Euler Angles
deg
ϕ
θ
ψ
Body Velocity
m/s
u
v
w
Time (s)
Position
m
N
E
D
Time (s)

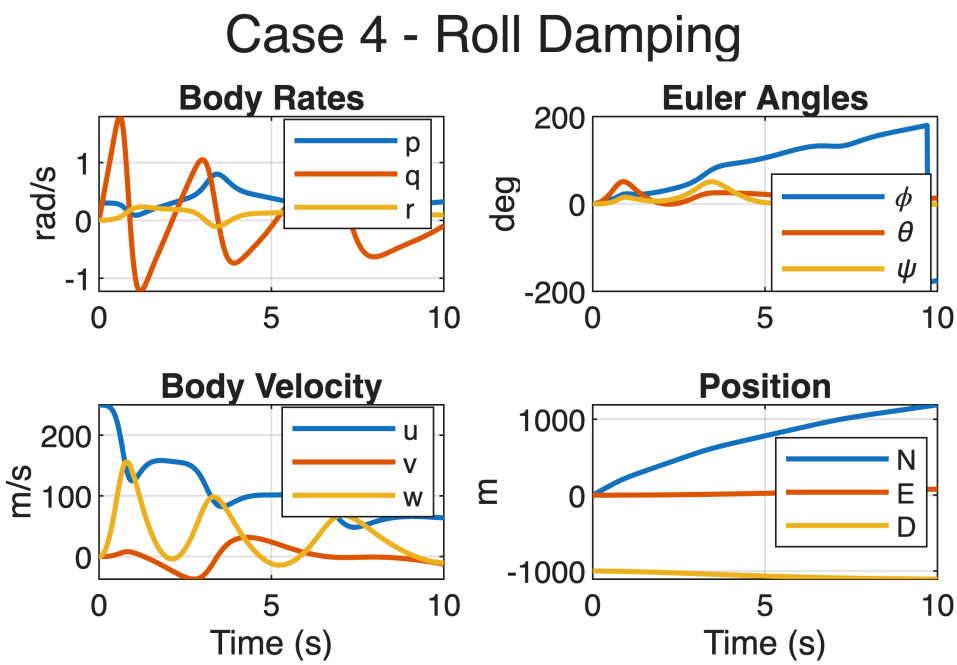

Case 3 - ForceField - Trajectory
Down [m]
East [m]
North [m]

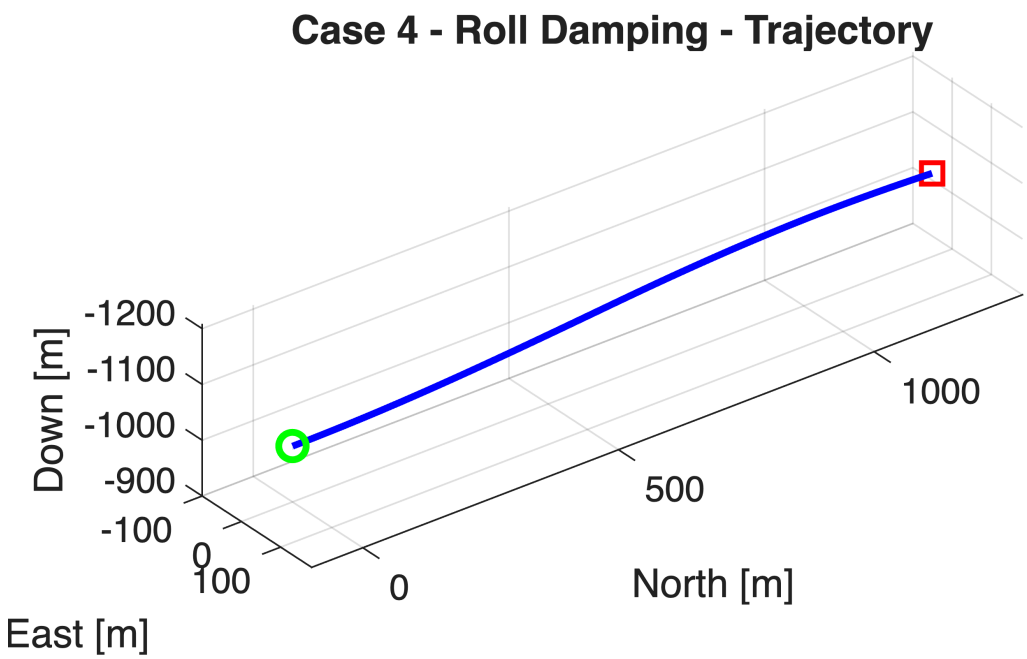


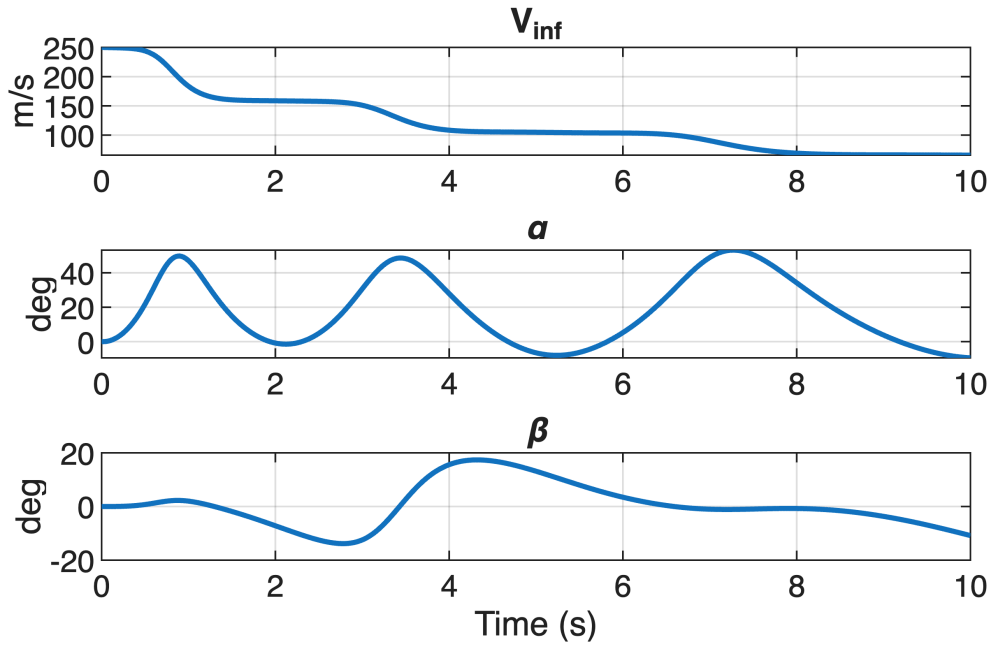

```
Drops due to gravity being stronger than the external force
```

```
run_flight_test(TC4, Mass, CG, J, wing, tailL, tailR, prop);
```

```
--------------------------------------------------
Case 4 - Roll Damping
--------------------------------------------------
  Model:             Brown_Full_Sim
  Simulation Time:   10 s
  Initial Position:  [0  0  -1000] m
  Initial Velocity:  [250  0  0] m/s
  Initial Euler:     [0  0  0] deg
  Initial Omega:     [0.3  0  0] rad/s
  Gravity:           [0  0  0] m/s^2
  Fext_B:            [0  0  0] N
  Mext_B:            [0  0  0] N-m
```


Case 4 - Roll Damping
Body Rates
rad/s
p
q
r
Euler Angles
deg
ϕ
θ
ψ
Body Velocity
m/s
u
v
w
Position
m
N
E
D
Time (s)
Time (s)

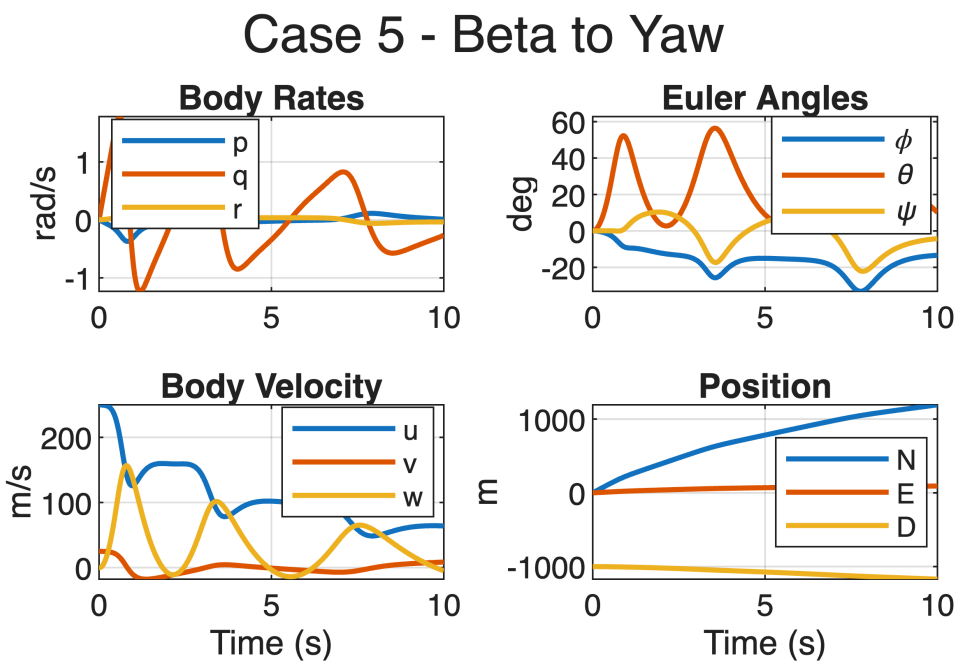

Case 4 - Roll Damping - Trajectory
Down [m]
North [m]
East [m]

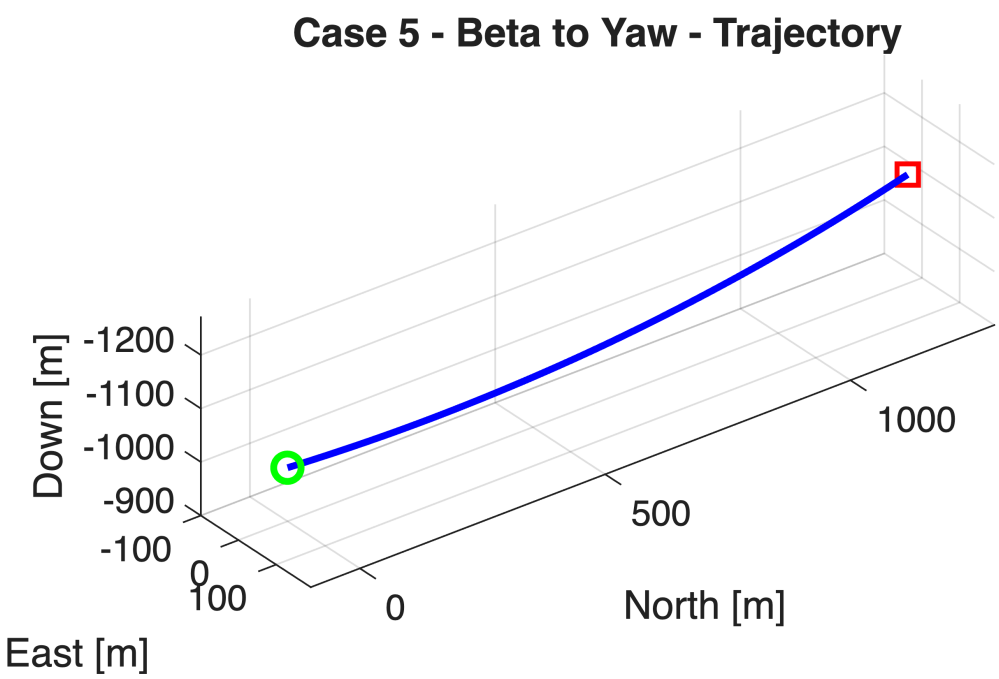


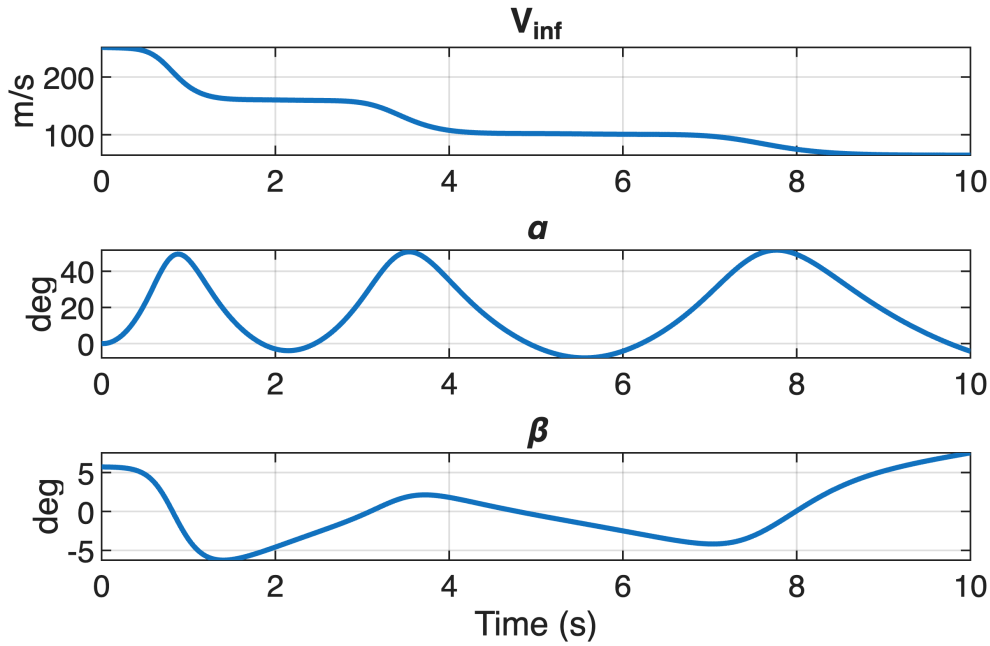

```
Stability seems to be different from prop plane likely due to V tail /nSeems like ti has a stability simmi
```

```
run_flight_test(TC5, Mass, CG, J, wing, tailL, tailR, prop);
```

```
--------------------------------------------------
Case 5 - Beta to Yaw
--------------------------------------------------
  Model:             Brown_Full_Sim
  Simulation Time:   10 s
  Initial Position:  [0  0  -1000] m
  Initial Velocity:  [250  25  0] m/s
  Initial Euler:     [0  0  0] deg
  Initial Omega:     [0  0  0] rad/s
  Gravity:           [0  0  0] m/s^2
  Fext_B:            [0  0  0] N
  Mext_B:            [0  0  0] N-m
```


Case 5 - Beta to Yaw
Body Rates
rad/s
p
q
r
Euler Angles
deg
ϕ
θ
ψ
Body Velocity
m/s
u
v
w
Time (s)
Position
m
N
E
D
Time (s)

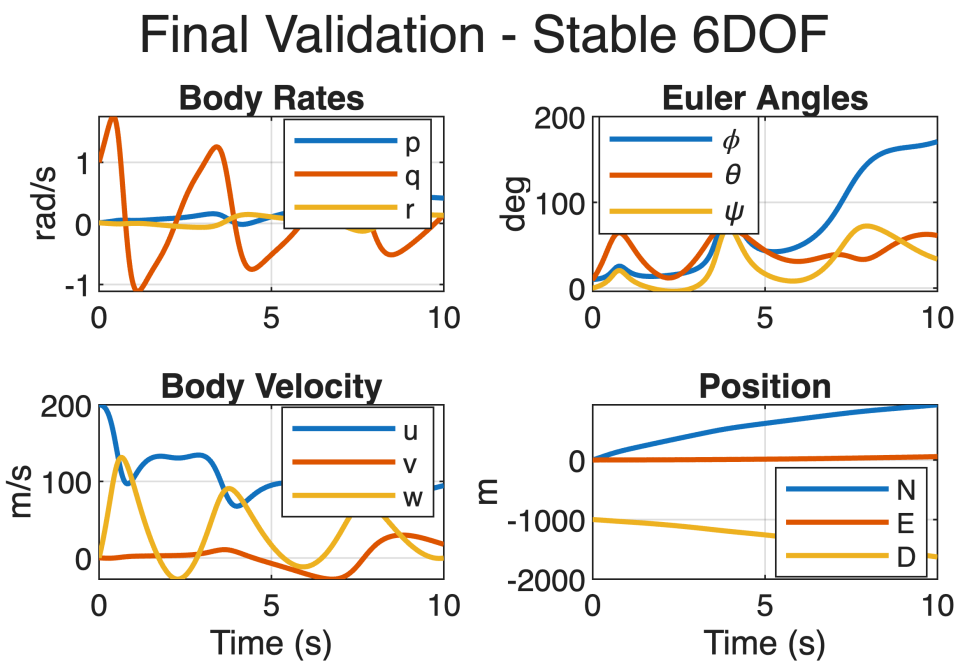

Case 5 - Beta to Yaw - Trajectory
Down [m]
-1200
-1100
-1000
-900
-100
0
100
East [m]
0
500
1000
North [m]

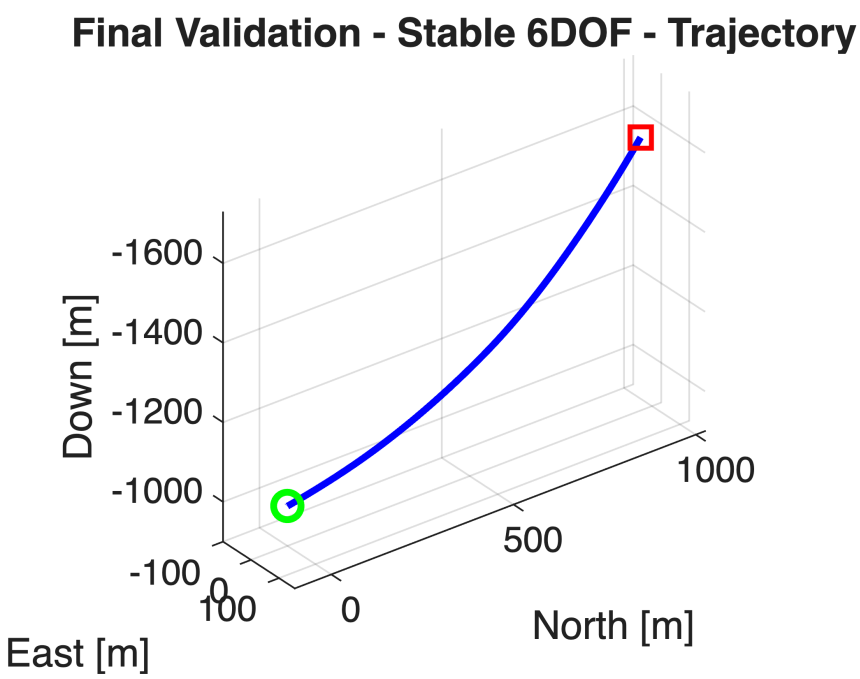


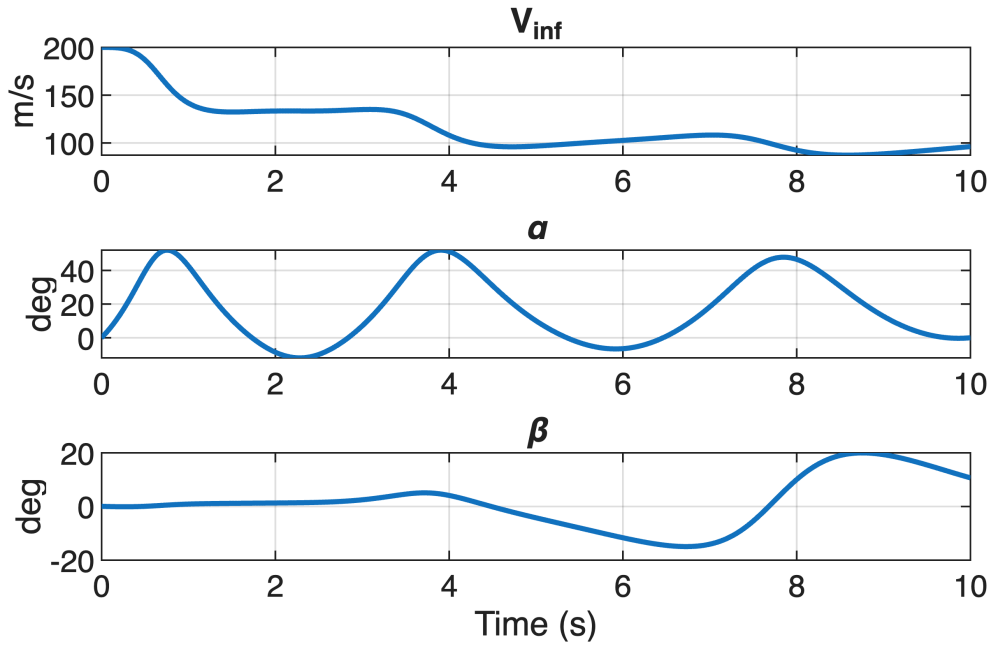
again lots of coupling in the axes that cant be simply explained due to V tial/n seeing the same paper ai

```
run_flight_test(TC6, Mass, CG, J, wing, tailL, tailR, prop);
```

```
--------------------------------------------------
Final Validation - Stable 6DOF
--------------------------------------------------
  Model:             Brown_Full_Sim
  Simulation Time:   10 s
  Initial Position:  [0  0  -1000] m
  Initial Velocity:  [200  0.25  0.25] m/s
  Initial Euler:     [10  10  0] deg
  Initial Omega:     [0.01  1  0.01] rad/s
  Gravity:           [0  0  -9.81] m/s^2
  Fext_B:            [0  0  0] N
  Mext_B:            [0  0  0] N-m
```


Final Validation - Stable 6DOF
Body Rates
rad/s
p
q
r
Euler Angles
deg
ϕ
θ
ψ
Body Velocity
m/s
u
v
w
Time (s)
Position
m
N
E
D
Time (s)


Final Validation - Stable 6DOF - Trajectory
Down [m]
East [m]
North [m]




```
Motors seem to bring plane upwards - Aircraft is much more complex than simple prop plane/n Will take a lo
```

Overall, the flight sim behaves as expected. However, the model is very different from the prop plane model given in class. Test cases need to be adapted to this aircraft-specific configuration, including mass, inertia, thrust vectors, and aero surfaces. Specifically, the V-tail is providing a significant challenge in terms of different stability dynamics. That will be investigated in the next week.

This report validates that the physics engine is correct, but also brings up some critical needs for effective test cases, as well as reveals some natural modes of motion.

Otherwise, I was able to get these two functions to open FlightGear; however, it's finicky because FlightGear is downloaded in a weird path on my computer, and so I get errors about where the aircraft data is located. I haven't been able to look around the aircraft yet, so there's still a lot of problems that need to be solved. I was able to load the loop-de-loop scenario too, and that's visually what happened in FlightGear, so I can confirm it worked. I don't know how to screenshot a video and put it in, but this is a screenshot clip from that.

```
addpath("Flight_Gear");
```

```
Warning: Name is nonexistent or not a directory: /Users/zbrown/Documents/Fifth year Spring/
Flight_Gear
```

```
cfg = fg_config();
fg_replay_from_out(out, cfg);
```

```
Warning: The aircraft geometry was not found at "/Users/zbrown/Library/Application Support/
FlightGear/fgdata_2024_1/Contents/Resources/data/Aircraft" on this computer. Make sure you copy it
before starting FlightGear.
```

Local functions

```
%% Local functions
function [Mass, CG, J, wing, tailL, tailR, prop, compData, tilt_angle] =
init_from_full_sim()
flight_mode = 0;
tilt_angle = 90 * flight_mode;
rho = 1.225;

prop.hub_offset = 0.50;
prop.k_Thrust = 1.2e-4;
prop.k_Torque = 1.5e-5;
prop.k_torque = 1.5e-5;

compData = {
    'Fuselage',   'fuselage', 1500, [7.50, 2.00, 1.60], [ 0.00,  0.00,
0.00], [0, 0, 0];
    'Ballast',    'box',      250,  [0.60, 0.60, 0.60], [ 3.00,  0.00,
0.00], [0, 0, 0];
    'Main Wing',  'box',      450,  [1.50, 14.50, 0.20], [-0.50,  0.00,
-0.60], [0, 0, 0];
```

```
    'L V-Tail',   'box',      90,   [1.40,  2.00, 0.08], [-3.50, -0.71,
-0.91], [ 45, 0, 0];
    'R V-Tail',   'box',      90,   [1.40,  2.00, 0.08], [-3.50,  0.71,
-0.91], [-45, 0, 0];
    'Boom L1',    'box',      55,   [4.50, 0.20, 0.20], [-0.20, -1.80,
-0.45], [0, 0, 0];
    'Boom L2',    'box',      55,   [4.50, 0.20, 0.20], [-0.20, -4.20,
-0.45], [0, 0, 0];
    'Boom L3',    'box',      55,   [4.50, 0.20, 0.20], [-0.20, -6.50,
-0.45], [0, 0, 0];
    'Boom R1',    'box',      55,   [4.50, 0.20, 0.20], [-0.20,  1.80,
-0.45], [0, 0, 0];
    'Boom R2',    'box',      55,   [4.50, 0.20, 0.20], [-0.20,  4.20,
-0.45], [0, 0, 0];
    'Boom R3',    'box',      55,   [4.50, 0.20, 0.20], [-0.20,  6.50,
-0.45], [0, 0, 0];
    'F-Arm L1',   'box',      5,    [0.40, 0.20, 0.20], [ 2.15, -1.80,
-0.55], [0, tilt_angle, 0];
    'F-Arm L2',   'box',      5,    [0.40, 0.20, 0.20], [ 2.15, -4.20,
-0.55], [0, tilt_angle, 0];
    'F-Arm L3',   'box',      5,    [0.40, 0.20, 0.20], [ 2.15, -6.50,
-0.55], [0, tilt_angle, 0];
    'F-Arm R1',   'box',      5,    [0.40, 0.20, 0.20], [ 2.15,  1.80,
-0.55], [0, tilt_angle, 0];
    'F-Arm R2',   'box',      5,    [0.40, 0.20, 0.20], [ 2.15,  4.20,
-0.55], [0, tilt_angle, 0];
    'F-Arm R3',   'box',      5,    [0.40, 0.20, 0.20], [ 2.15,  6.50,
-0.55], [0, tilt_angle, 0];
    'F-Rotor L1', 'crossprop',25,   [1.60, 0.15, 0.05], [ 1.95, -1.80,
-0.45], [0, tilt_angle, 0];
    'F-Rotor L2', 'crossprop',25,   [1.60, 0.15, 0.05], [ 1.95, -4.20,
-0.45], [0, tilt_angle, 0];
    'F-Rotor L3', 'crossprop',25,   [1.60, 0.15, 0.05], [ 1.95, -6.50,
-0.45], [0, tilt_angle, 0];
    'F-Rotor R1', 'crossprop',25,   [1.60, 0.15, 0.05], [ 1.95,  1.80,
-0.45], [0, tilt_angle, 0];
    'F-Rotor R2', 'crossprop',25,   [1.60, 0.15, 0.05], [ 1.95,  4.20,
-0.45], [0, tilt_angle, 0];
    'F-Rotor R3', 'crossprop',25,   [1.60, 0.15, 0.05], [ 1.95,  6.50,
-0.45], [0, tilt_angle, 0];
    'R-Rotor L1', 'crossprop',25,   [1.60, 0.15, 0.05], [-2.20, -1.80,
-0.65], [0, 0, 0];
    'R-Rotor L2', 'crossprop',25,   [1.60, 0.15, 0.05], [-2.20, -4.20,
-0.65], [0, 0, 0];
    'R-Rotor L3', 'crossprop',25,   [1.60, 0.15, 0.05], [-2.20, -6.50,
-0.65], [0, 0, 0];
    'R-Rotor R1', 'crossprop',25,   [1.60, 0.15, 0.05], [-2.20,  1.80,
-0.65], [0, 0, 0];
    'R-Rotor R2', 'crossprop',25,   [1.60, 0.15, 0.05], [-2.20,  4.20,
-0.65], [0, 0, 0];
```

```
    'R-Rotor R3', 'crossprop',25,   [1.60, 0.15, 0.05], [-2.20,  6.50,
-0.65], [0, 0, 0];
};

% format: {name, normal_vec, CL0, e, i, CD0, CDa, a0, CM0, CMa, CLa}
aeroData = {
    'Main Wing', [0; 0; -1],                 0.05, 0.9, 0.05, 0.01, 1,
0.05, -0.05, 0, 0.01;
    'L V-Tail',  [0; sind(45); -cosd(45)],   0,    0.8, 0,    0.01, 1,
0,     0,    0, 0.01;
    'R V-Tail',  [0; -sind(45); -cosd(45)],  0,    0.8, 0,    0.01, 1,
0,     0,    0, 0.01;
};

N = size(compData, 1);

Mtot = 0;
CG_row = [0, 0, 0];
for i = 1:N
    m = compData{i, 3};
    pos = compData{i, 5};
    Mtot = Mtot + m;
    CG_row = CG_row + m * pos;
end
CG_row = CG_row / Mtot;

I_total = zeros(3,3);
for i = 1:N
    type = compData{i, 2};
    m = compData{i, 3};
    dim = compData{i, 4};
    pos = compData{i, 5};
    eul = compData{i, 6} * (pi/180);

    if strcmp(type, 'crossprop')
        D = dim(1); W = dim(2); T = dim(3);
        Ixx1 = (1/12)*(m/2)*(W^2 + T^2);
        Iyy1 = (1/12)*(m/2)*(D^2 + T^2);
        Izz1 = (1/12)*(m/2)*(D^2 + W^2);
        Ixx2 = (1/12)*(m/2)*(D^2 + T^2);
        Iyy2 = (1/12)*(m/2)*(W^2 + T^2);
        Izz2 = (1/12)*(m/2)*(D^2 + W^2);
        I_local = diag([Ixx1+Ixx2, Iyy1+Iyy2, Izz1+Izz2]);
    else
        xL = dim(1); yL = abs(dim(2)); zL = dim(3);
        I_local = diag([(1/12)*m*(yL^2 + zL^2), (1/12)*m*(xL^2 + zL^2),
(1/12)*m*(xL^2 + yL^2)]);
    end

    phi = eul(1); theta = eul(2); psi = eul(3);
```

```
    Rx = [1 0 0; 0 cos(phi) -sin(phi); 0 sin(phi) cos(phi)];
    Ry = [cos(theta) 0 sin(theta); 0 1 0; -sin(theta) 0 cos(theta)];
    Rz = [cos(psi) -sin(psi) 0; sin(psi) cos(psi) 0; 0 0 1];
    R = Rz * Ry * Rx;

    I_rot = R * I_local * R';
    d = (pos - CG_row)';
    I_total = I_total + I_rot + m * ((d'*d)*eye(3) - (d*d'));
end

Mass = Mtot;
CG = CG_row';
J = I_total;

wing.c = compData{3, 4}(1); wing.b = compData{3, 4}(2); wing.S = wing.b *
wing.c;
wing.half_rho_S = 0.5 * rho * wing.S; wing.pos = (compData{3, 5})';
wing.n = aeroData{1, 2}; wing.CL0 = aeroData{1, 3}; wing.e = aeroData{1, 4};
wing.i = aeroData{1, 5}; wing.CD0 = aeroData{1, 6}; wing.CDa = aeroData{1,
7};
wing.a0 = aeroData{1, 8}; wing.CM0 = aeroData{1, 9}; wing.CMa = aeroData{1,
10};
wing.CLa = aeroData{1, 11}; wing.AR = wing.b / wing.c;

tailL.c = compData{4, 4}(1); tailL.b = compData{4, 4}(2); tailL.S = tailL.b
* tailL.c;
tailL.half_rho_S = 0.5 * rho * tailL.S; tailL.pos = (compData{4, 5})';
tailL.n = aeroData{2, 2}; tailL.CL0 = aeroData{2, 3}; tailL.e = aeroData{2,
4};
tailL.i = aeroData{2, 5}; tailL.CD0 = aeroData{2, 6}; tailL.CDa =
aeroData{2, 7};
tailL.a0 = aeroData{2, 8}; tailL.CM0 = aeroData{2, 9}; tailL.CMa =
aeroData{2, 10};
tailL.CLa = aeroData{2, 11}; tailL.AR = tailL.b / tailL.c;

tailR.c = compData{5, 4}(1); tailR.b = compData{5, 4}(2); tailR.S = tailR.b
* tailR.c;
tailR.half_rho_S = 0.5 * rho * tailR.S; tailR.pos = (compData{5, 5})';
tailR.n = aeroData{3, 2}; tailR.CL0 = aeroData{3, 3}; tailR.e = aeroData{3,
4};
tailR.i = aeroData{3, 5}; tailR.CD0 = aeroData{3, 6}; tailR.CDa =
aeroData{3, 7};
tailR.a0 = aeroData{3, 8}; tailR.CM0 = aeroData{3, 9}; tailR.CMa =
aeroData{3, 10};
tailR.CLa = aeroData{3, 11}; tailR.AR = tailR.b / tailR.c;

idx_FR = find(contains(compData(:,1), 'F-Rotor R'));
idx_FL = find(contains(compData(:,1), 'F-Rotor L'));
idx_RR = find(contains(compData(:,1), 'R-Rotor R'));
idx_RL = find(contains(compData(:,1), 'R-Rotor L'));
```

```
prop.posFR = cell2mat(compData(idx_FR, 5));
prop.posFL = cell2mat(compData(idx_FL, 5));
prop.posRR = cell2mat(compData(idx_RR, 5));
prop.posRL = cell2mat(compData(idx_RL, 5));
prop.Lspin_dir = [-1; 1; -1];
prop.Rspin_dir = [1; -1; 1];
end

function out = run_flight_test(Case, Mass, CG, J, wing, tailL, tailR, prop)
if isscalar(Case.g)
    g_vec = [0;0;Case.g];
else
    g_vec = Case.g(:);
end
disp('----------------------------------------------------');
disp(Case.name);
disp('----------------------------------------------------');
fprintf('  Model:            %s\n', Case.model);
fprintf('  Simulation Time:  %g s\n', Case.t_stop);
fprintf('  Initial Position: [%g  %g  %g] m\n', Case.pos_init);
fprintf('  Initial Velocity: [%g  %g  %g] m/s\n', Case.V_init);
fprintf('  Initial Euler:    [%g  %g  %g] deg\n', Case.eul_init_deg);
fprintf('  Initial Omega:    [%g  %g  %g] rad/s\n', Case.omega_init);
fprintf('  Gravity:          [%g  %g  %g] m/s^2\n', g_vec);
fprintf('  Fext_B:           [%g  %g  %g] N\n', Case.Fext_B);
fprintf('  Mext_B:           [%g  %g  %g] N-m\n', Case.Mext_B);
disp(' ');


assignin('base', 'Mass', Mass);
assignin('base', 'm', Mass);
assignin('base', 'CG', CG);
assignin('base', 'J', J);
assignin('base', 'wing', wing);
assignin('base', 'tailL', tailL);
assignin('base', 'tailR', tailR);
assignin('base', 'prop', prop);

assignin('base', 'g', g_vec);
assignin('base', 'pos_init', Case.pos_init);
assignin('base', 'V_init', Case.V_init);
assignin('base', 'eul_init', deg2rad(Case.eul_init_deg(:)));
assignin('base', 'omega_init', Case.omega_init);
assignin('base', 'Fext_B', Case.Fext_B);
assignin('base', 'Mext_B', Case.Mext_B);
assignin('base', 'Motor_RPMs', Case.Motor_RPMs);
assignin('base', 'Tilt_angles', Case.Tilt_angles);
```

```
out = sim(Case.model, 'StopTime', num2str(Case.t_stop),
'ReturnWorkspaceOutputs', 'on');

[omega, t_omega] = get_ts_data(out, 'omega');
[eul, t_eul] = get_ts_data(out, 'eul');
[v_b, t_v_b] = get_ts_data(out, 'v_b');
[position, t_pos] = get_ts_data(out, 'position');
[vinf, t_vinf] = get_ts_data(out, 'vinf');
[alpha, t_alpha] = get_ts_data(out, 'alpha');
[beta, t_beta] = get_ts_data(out, 'beta');

if ~isempty(eul)
    eul = rad2deg(eul);
end
if ~isempty(alpha)
    alpha = rad2deg(alpha);
end
if ~isempty(beta)
    beta = rad2deg(beta);
end

figure('Name', Case.name);
sgtitle(Case.name);

subplot(2,2,1);
plot(t_omega, omega, 'LineWidth', 1.5); grid on;
title('Body Rates'); ylabel('rad/s'); legend('p','q','r','Location','best');

subplot(2,2,2);
plot(t_eul, eul, 'LineWidth', 1.5); grid on;
title('Euler Angles'); ylabel('deg');
legend('\phi','\theta','\psi','Location','best');

subplot(2,2,3);
plot(t_v_b, v_b, 'LineWidth', 1.5); grid on;
title('Body Velocity'); ylabel('m/s'); xlabel('Time (s)');
legend('u','v','w','Location','best');

subplot(2,2,4);
plot(t_pos, position, 'LineWidth', 1.5); grid on;
title('Position'); ylabel('m'); xlabel('Time (s)');
legend('N','E','D','Location','best');

if ~isempty(position)
    X = position(:,1); Y = position(:,2); Z = position(:,3);

    figure('Name', [Case.name, ' Trajectory']);
    plot3(X, Y, Z, 'b-', 'LineWidth', 2); grid on; hold on;
    set(gca, 'ZDir', 'reverse', 'YDir', 'reverse');
    plot3(X(1), Y(1), Z(1), 'go', 'MarkerSize', 8, 'LineWidth', 2);
```

```
    plot3(X(end), Y(end), Z(end), 'rs', 'MarkerSize', 8, 'LineWidth', 2);
    xlabel('North [m]'); ylabel('East [m]'); zlabel('Down [m]');

    title([Case.name, ' - Trajectory']);
    axis equal;
    xlim([min(X)-100,max(X)+100]);
    ylim([min(Y)-100,max(Y)+100]);
    zlim([min(Z)-100,max(Z)+100]);
end

if ~isempty(vinf) || ~isempty(alpha) || ~isempty(beta)
    figure('Name', [Case.name, ' Air Data']);

    subplot(3,1,1);
    plot(t_vinf, vinf, 'LineWidth', 1.5); grid on;
    title('V_{inf}'); ylabel('m/s');

    subplot(3,1,2);
    plot(t_alpha, alpha, 'LineWidth', 1.5); grid on;
    title('\alpha'); ylabel('deg');

    subplot(3,1,3);
    plot(t_beta, beta, 'LineWidth', 1.5); grid on;
    title('\beta'); ylabel('deg'); xlabel('Time (s)');
end
disp(Case.note);
end

function [data, t] = get_ts_data(out, var_name)
data = [];
t = [];

try
    ts = out.get(var_name);
catch
    return;
end

if isempty(ts)
    return;
end

t = ts.Time;
data = squeeze(ts.Data);

if isempty(data)
    return;
end

if isvector(data)
```

```
    data = data(:);
elseif size(data,1) <= 3 && size(data,2) > size(data,1)
    data = data';
end
end
```